Cosmological Implications of Arithmetic Spacetime: If the Universe Lives on Spec($\mathbb{Z}$)

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Abstract

The arithmetic vacuum engineering program rests on a foundational premise: the quantum vacuum has the arithmetic structure of Spec($\mathbb{Z}$), the scheme of integers. We examine the cosmological implications of this premise. If spacetime is fundamentally arithmetic, then: (1) the cosmological constant equals $\zeta(-3)/120$ in Planck units, potentially resolving the cosmological constant problem by replacing cutoff regularization with ζ-regularization as the *correct* description; (2) the Big Bang is a phase transition of the Bost-Connes system at $\beta = 1$, where the Galois symmetry $\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q})$ spontaneously breaks, differentiating the primes; (3) dark energy is the Casimir energy of Spec($\mathbb{Z}$) itself, i.e., the zero-point energy of the arithmetic vacuum; (4) the fine structure constant and other dimensionless constants are determined by the arithmetic geometry of Spec($\mathbb{Z}$) (zeta values, class numbers, regulator volumes); (5) the multiverse, if it exists, is the set of localizations $\mathrm{Spec}(\mathbb{Z}[S^{-1}])$ for different sets S of primes, each defining a universe with different physical laws via topos logic change. We formulate each implication as a precise conjecture and identify the observational signatures that would distinguish arithmetic spacetime from conventional cosmology.

1 Introduction

The arithmetic vacuum engineering program [1, 2, 3] is built on a premise that is rarely stated explicitly:

Conjecture 1 (Arithmetic spacetime)**.** *The fundamental structure of spacetime is the arithmetic scheme* Spec($\mathbb{Z}$)*. The quantum vacuum is a sheaf on this scheme. The Riemann zeta function* $\zeta(s)$*, as the zeta function of* Spec($\mathbb{Z}$)*, encodes the vacuum's physical properties.*

This paper does not prove the conjecture. Instead, we ask: *if it is true, what are the consequences for cosmology?*

The results are striking. Several of the deepest puzzles in modern cosmology — the cosmological constant problem, the origin of the Big Bang, the nature of dark energy, the values of fundamental constants — acquire natural (if speculative) explanations within the arithmetic framework.

2 The cosmological constant problem

2.1 The problem

The cosmological constant Λ is the energy density of the vacuum. Quantum field theory predicts

$$\rho_{\mathrm{vac}}^{\mathrm{QFT}} \sim \frac{\hbar c}{l_P^4} \sim 10^{113}\,\mathrm{J/m^3}, \tag{1}$$

while observation gives $\rho_{\mathrm{vac}}^{\mathrm{obs}} \sim 10^{-9}\,\mathrm{J/m^3}$. The discrepancy of $\sim 10^{122}$ is the worst prediction in the history of physics [4].

2.2 The arithmetic resolution

The QFT prediction uses *cutoff regularization*: $\rho \sim \int_0^{E_P} \omega^3 \, d\omega \sim E_P^4$. This diverges and must be cut off at the Planck scale.

Zeta regularization replaces this with

$$\rho_{\text{vac}}^{\zeta} \propto \zeta(-3) = \frac{1}{120}. \tag{2}$$

Conjecture 2 (Arithmetic cosmological constant)**.** *The physical cosmological constant is*

$$\Lambda = \frac{8\pi G}{c^4} \cdot \frac{E_P}{l_P^3} \cdot \zeta(-3) = \frac{8\pi}{120} \, l_P^{-2}, \tag{3}$$

where the factor $\zeta(-3) = 1/120$ replaces the divergent cutoff integral.

This gives $\Lambda \sim 0.2 \, l_P^{-2}$, which is still far from the observed value $\Lambda_{\text{obs}} \sim 10^{-122} \, l_P^{-2}$. However, the key point is not the numerical value but the *framework*: if the vacuum is $\text{Spec}(\mathbb{Z})$, then the cosmological constant is determined by $\zeta(-3)$, not by a cutoff. The remaining discrepancy may be explained by the *partial muting* of primes in the actual vacuum (see Section 4).

3 The Big Bang as Bost-Connes phase transition

The Bost-Connes system has a phase transition at $\beta = 1$:

- $\beta < 1$ (high temperature): unique KMS state. Full symmetry $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$ is preserved. All primes are indistinguishable.
- $\beta > 1$ (low temperature): the symmetry spontaneously breaks. KMS states form a continuous family parameterized by $\hat{\mathbb{Z}}^* = \prod_p \mathbb{Z}_p^*$. Primes become individually addressable.

Conjecture 3 (Arithmetic Big Bang)**.** *The Big Bang is the Bost-Connes phase transition at $\beta = 1$.*

Before the transition ($\beta < 1$): the universe is in the symmetric phase. No individual primes exist. No particle species are distinguishable (because the Standard Model requires the broken phase of the internal space F). Spacetime has maximal symmetry.

At the transition ($\beta = 1$): the Galois symmetry breaks. Primes differentiate. The Euler product structure of $\zeta(s)$ emerges. Particle species appear (via Connes' spectral action on $M^4 \times F$). The arrow of time appears (because the broken phase selects a specific KMS state).

After the transition ($\beta > 1$): the universe we observe. Each prime p contributes its Euler factor $(1 - p^{-\beta})^{-1}$ to the vacuum structure.

Observational signature. The cosmic microwave background (CMB) encodes fluctuations from the epoch near the phase transition. If the Big Bang is a BC transition, the primordial power spectrum should reflect the number-theoretic structure of $\zeta(s)$ near $\beta = 1$:

$$P(k) \propto \left| \frac{\zeta(1 + \epsilon + ik)}{\zeta(1 + \epsilon)} \right|^2 \tag{4}$$

for small $\epsilon > 0$ (proximity to the critical point). The Riemann zeros on the critical line would imprint as oscillatory features in the CMB power spectrum.

4 Dark energy as arithmetic Casimir energy

Conjecture 4 (Arithmetic dark energy)**.** *Dark energy is the Casimir energy of* $\mathrm{Spec}(\mathbb{Z})$ *itself: the zero-point energy of the vacuum computed on the arithmetic scheme, with boundary conditions set by the large-scale topology of the universe.*

In standard Casimir physics, the vacuum energy depends on boundary conditions (plate separation, geometry). On $\mathrm{Spec}(\mathbb{Z})$, the "boundaries" are:

- The archimedean place $v = \infty$ (real numbers, $\mathbb{R}$): large-scale classical spacetime.
- The non-archimedean places $v = p$ (p-adic numbers, $\mathbb{Q}_p$): the arithmetic microstructure.

The adelic product formula $\prod_v |x|_v = 1$ constrains the total vacuum energy. If certain primes are "partially muted" by cosmological evolution (e.g., by the dilution of p-adic modes during inflation), the effective vacuum energy would be

$$\rho_{\mathrm{dark}} \propto \zeta(-3) \cdot \prod_{p \in S_{\mathrm{muted}}} (1 - p^3), \tag{5}$$

which can take a very small value if the muted prime set S_{muted} is chosen appropriately.

The observed dark energy density $\rho_{\mathrm{dark}} \sim 10^{-9}\,\mathrm{J/m^3}$ constrains the muted prime set. This is a quantitative prediction of the arithmetic framework that could, in principle, be tested.

5 Fundamental constants from arithmetic geometry

If spacetime is $\mathrm{Spec}(\mathbb{Z})$, then dimensionless physical constants should be computable from its arithmetic-geometric invariants.

Conjecture 5 (Arithmetic constants)**.** *The dimensionless constants of physics (fine structure constant α, weak mixing angle θ_W, Yukawa couplings, etc.) are determined by the arithmetic geometry of* $\mathrm{Spec}(\mathbb{Z})$*: zeta special values, K-group orders, regulator volumes, and class numbers of number fields.*

Partial evidence from Connes' NCG Standard Model [5]: the spectral action ($f(D/\Lambda)$) on $M^4 \times F$ determines the Standard Model coupling constants at unification scale as functions of the parameters of F. If F has BC structure (Conjecture 1), these parameters are arithmetic invariants.

Specifically:

- $\zeta(-1) = -1/12$ appears in the 1-loop vacuum energy, related to the trace anomaly and conformal symmetry breaking.
- $\zeta(-3) = 1/120$ determines the 3D Casimir energy, related to the cosmological constant.
- $\zeta(2) = \pi^2/6$ and $\zeta(4) = \pi^4/90$ appear in black-body radiation (Stefan-Boltzmann law), connecting thermodynamics to ζ values.
- $K_2(\mathbb{Z}) = \mathbb{Z}/2$ (Milnor) may encode the discrete symmetries (C, P, T) of the Standard Model.

6 The multiverse as localization

If our universe corresponds to $\mathrm{Spec}(\mathbb{Z})$, what corresponds to other possible universes?

Conjecture 6 (Arithmetic multiverse)**.** *The "multiverse" is the set of localizations of $\mathbb{Z}$:*

$$\mathcal{M} = \{\mathrm{Spec}(\mathbb{Z}[S^{-1}]) : S \subseteq \mathrm{Primes}\}. \tag{6}$$

Each choice of S (a set of primes to "mute") defines a universe with a different étale topos, different internal logic, and therefore different physical laws.

The extreme cases:

- $S = \emptyset$: $\mathrm{Spec}(\mathbb{Z})$. Our universe. All primes active. Full WEC. Standard physics.
- $S = \{p\}$: $\mathrm{Spec}(\mathbb{Z}[1/p])$. A universe missing one prime. WEC violated for p. Warp drive possible.
- $S =$ all primes: $\mathrm{Spec}(\mathbb{Q})$. A universe with no primes. All Euler factors removed. $\zeta_S(s) = 1$ (trivial). Zero vacuum energy. No particles, no forces, no structure.

This provides a natural ordering of universes: more primes muted $\Rightarrow$ simpler vacuum structure $\Rightarrow$ fewer physical phenomena. Our universe ($S = \emptyset$) is the "maximal" universe, with all primes active and the richest possible physics.

The anthropic principle, arithmetized. The question "why does our universe have these physical laws?" becomes "why is $S = \emptyset$?" — or equivalently, "why are all primes active?" A possible answer: $S = \emptyset$ is the only choice that supports sufficient complexity for observers (since muting primes removes degrees of freedom needed for chemistry and biology).

7 Observational tests

Table 1: Observational signatures of arithmetic spacetime.

Prediction	Observable	Current status
$\Lambda = \zeta(-3) \times$ (Planck)	Cosmological constant	Discrepant by $\sim 10^{120}$; may be resolved by partial muting
BC phase transition at Big Bang	CMB power spectrum with ζ-function features	Not yet searched for
Dark energy $\propto \prod(1 - p^3)$	Equation of state $w(z)$ evolution	Consistent with $w = -1$ (current data)
Constants from ζ values	Predictions for α, θ_W at unification	Requires explicit computation within NCG
Riemann zeros in CMB	Oscillations at positions of ζ zeros	Tantalizing claims exist [10] but unconfirmed

The most immediately testable prediction is the CMB signature of the BC phase transition: oscillatory features in the primordial power spectrum correlated with the Riemann zeros. This can be searched for in existing Planck satellite data.

8 Discussion

8.1 Falsifiability

Each conjecture in this paper is, in principle, falsifiable:

- Conjecture 2: if the cosmological constant cannot be expressed in terms of ζ values and arithmetic invariants, the premise fails.
- Conjecture 3: if the CMB shows no trace of ζ-function structure, the BC phase transition interpretation is disfavored.
- Conjecture 5: if the NCG spectral action fails to predict any measured constant, the arithmetic program loses predictive power.

8.2 Relation to other approaches

The arithmetic spacetime hypothesis intersects with several established research programs:

- **p-adic cosmology** (Volovich 1987, Dragovich [7]): uses p-adic numbers in cosmological models. Our framework provides a unified (adelic) setting.
- **Spectral geometry** (Connes [6]): derives physics from spectral data. We add the arithmetic ($\mathrm{Spec}(\mathbb{Z})$) perspective.
- **Amplituhedron / motivic periods** (Arkani-Hamed [8], Brown [9]): scattering amplitudes involve periods of mixed motives. Our framework connects these to the vacuum structure via the motivic cohomology of $\mathrm{Spec}(\mathbb{Z})$.

9 Conclusion

If spacetime is $\mathrm{Spec}(\mathbb{Z})$, then:

1. The cosmological constant is $\zeta(-3)$ in natural units, not a cutoff artifact.
2. The Big Bang is a Bost-Connes phase transition where primes differentiate.
3. Dark energy is the Casimir energy of the arithmetic scheme.
4. Fundamental constants are arithmetic-geometric invariants.
5. The multiverse is the lattice of localizations of $\mathbb{Z}$.

None of these are proven. All are falsifiable. Together, they form a coherent cosmological narrative in which number theory is not an abstract mathematical discipline but the science of the universe's fundamental structure.

The experimental program of [1] — testing the Euler product structure of vacuum energy in a superconducting circuit — is the first step toward confirming or refuting this picture.

References

[1] Wright Brothers, companion paper A (2026).

[2] Wright Brothers, companion paper B (2026).

[3] Wright Brothers, companion paper C (2026).

[4] S. Weinberg, Rev. Mod. Phys. **61**, 1 (1989).

[5] A. H. Chamseddine, A. Connes, and M. Marcolli, Adv. Theor. Math. Phys. **11**, 991 (2007).

[6] A. Connes, Commun. Math. Phys. **182**, 155 (1996).

[7] B. Dragovich et al., p-Adic Numbers Ultrametric Anal. Appl. **9**, 87 (2017).

[8] N. Arkani-Hamed and J. Trnka, JHEP **2014**, 030 (2014).

[9] F. Brown, Commun. Math. Phys. **338**, 1 (2015).

[10] M. Wolf, arXiv:1002.1891 (2010).